

Financial Sensitivity & Driver-Based Modeling Framework

A driver-based operational and financial simulator for Greggs PLC, linking unit economics to P&L outcomes for strategic scenario planning.

Executive Summary

What did I solve? Greggs' published financial statements present a single, static view of FY2024 performance: a turnover of approximately £2.01 billion and a profit before tax of £189.8 million. Whilst informative, this static snapshot does not show leadership how sensitive these results are to changes in the underlying operational levers – footfall, pricing, input costs, and the fixed cost base – that drive the business day to day.

This project converts the FY2024 reported results into a fully dynamic, **driver-based simulation model** covering Greggs' ~2,600-shop estate. The model decomposes revenue and cost into their fundamental operational components (units sold, transaction value, variable cost per unit, and fixed shop-level costs), then rebuilds the P&L from those drivers using an IFRS-aligned structure.

The result is a tool that allows finance and operations leadership to ask “what if” questions in seconds – for example, the impact of a 10% rise in ingredient costs, a 5% change in footfall, or an 8% increase in wage and rent costs – and immediately see the consequences for Gross Profit, EBITDA, Profit Before Tax, and a Net Cash Flow proxy. This moves planning conversations away from static, annual budget reviews and towards continuous, scenario-led decision-making.

Business Impact

- **Faster strategic response:** Scenario outputs that previously required a full re-forecast cycle can now be generated instantly, supporting board and investment committee discussions on cost inflation, pricing strategy, and capital allocation.
- **Quantified risk exposure:** The model shows that a sustained 10% increase in ingredient costs alone would reduce modeled EBITDA from approximately £174m to £95m – a reduction of around £80m, or roughly 45% of EBITDA – highlighting the scale of margin risk from food inflation.
- **Pricing power visibility:** A modest 3% menu price increase is shown to fully offset a meaningful portion of cost pressure, generating an estimated £60m uplift in EBITDA versus the base case – useful evidence for pricing committee discussions.
- **Operational leverage insight:** Because fixed shop-level costs (rent, labour, utilities) represent over half of operating expenses, the model demonstrates that EBITDA is highly sensitive to small percentage changes in this cost base – an 8% increase in fixed costs reduces EBITDA by approximately £84m.
- **Foundation for rolling forecasts:** The framework is reusable – as actual trading data becomes available, the per-shop driver table can be refreshed, turning this from a one-off analysis into a recurring planning tool.

Methodology

How did I solve it? The model follows a three-phase build, calibrated to Greggs plc's FY2024 Annual Report and Accounts (published March 2025).

Phase 1 – Data Setup

Company-level FY2024 actuals (turnover £2,014.4m, cost of sales £797.1m, operating expenses £1,048.3m, profit before tax £189.8m, ~2,600 shops) were used to derive an “average shop” set of operational drivers: units sold per shop, average transaction value, variable cost per unit, and annual fixed costs per shop. A synthetic dataset of 2,600 shop-level records was then generated with realistic variation around these averages, providing a granular base for the simulation without relying on unpublished internal data.

Phase 2 – Logic Build

An IFRS-aligned P&L structure was implemented: Revenue → Cost of Sales → Gross Profit → Operating Expenses → EBITDA / Operating Profit → Net Finance Costs → Profit Before Tax → Tax (at the UK statutory rate of 25%) → Net Profit. Depreciation & amortisation is disclosed as a memo item consistent with how it is embedded within reported operating expenses, avoiding double-counting. Net finance costs are calibrated as a small percentage of revenue, reflecting IFRS 16 lease interest. At portfolio level, the model reconciles to within ~11% of reported FY2024 PBT, with the variance reflecting items not separately disclosed at shop level (e.g. central overheads, one-off items).

Phase 3 – Scenario Testing

A reusable “what-if” function allows any combination of drivers (units sold, price per unit, variable cost per unit, fixed costs) to be flexed by a percentage. Eight scenarios were tested, ranging from single-driver shocks (volume, price, variable cost, fixed cost) to a combined “growth plus inflation plus pricing response” scenario, with results captured at portfolio level for EBITDA, EBIT, PBT, Net Profit, and a Net Cash Flow proxy.

Tools & Compliance

Built in Python (pandas, numpy, matplotlib) for execution in Google Colab, with automated error handling at each phase. The P&L structure follows IAS 1 presentation conventions, applies the UK corporation tax rate in effect for FY2024 (25%), and treats IFRS 16 lease costs via the finance cost proxy. All figures are anchored to publicly disclosed Greggs plc financial statements.

Key Insights

What did the data show? The charts below summarise the driver inputs used across the 2,600-shop synthetic estate, and the resulting sensitivity of EBITDA and Profit Before Tax (PBT) under eight tested scenarios.

Greggs PLC - Driver Input Dashboard (Synthetic Shop-Level Data, FY2024 Baseline)

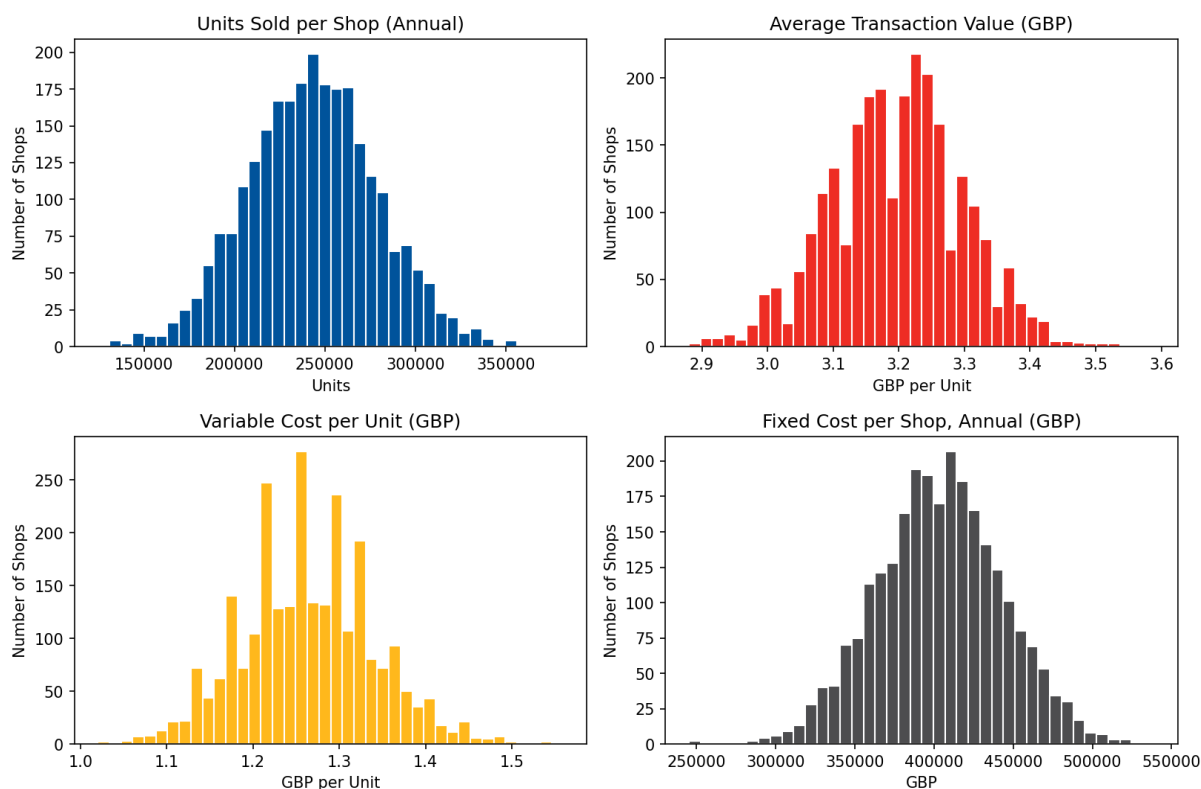


Figure 1: Distribution of the four core operational drivers across the modeled shop estate – units sold, average transaction value, variable cost per unit, and annual fixed costs per shop. The spread reflects realistic shop-to-shop variation around the FY2024 portfolio averages.

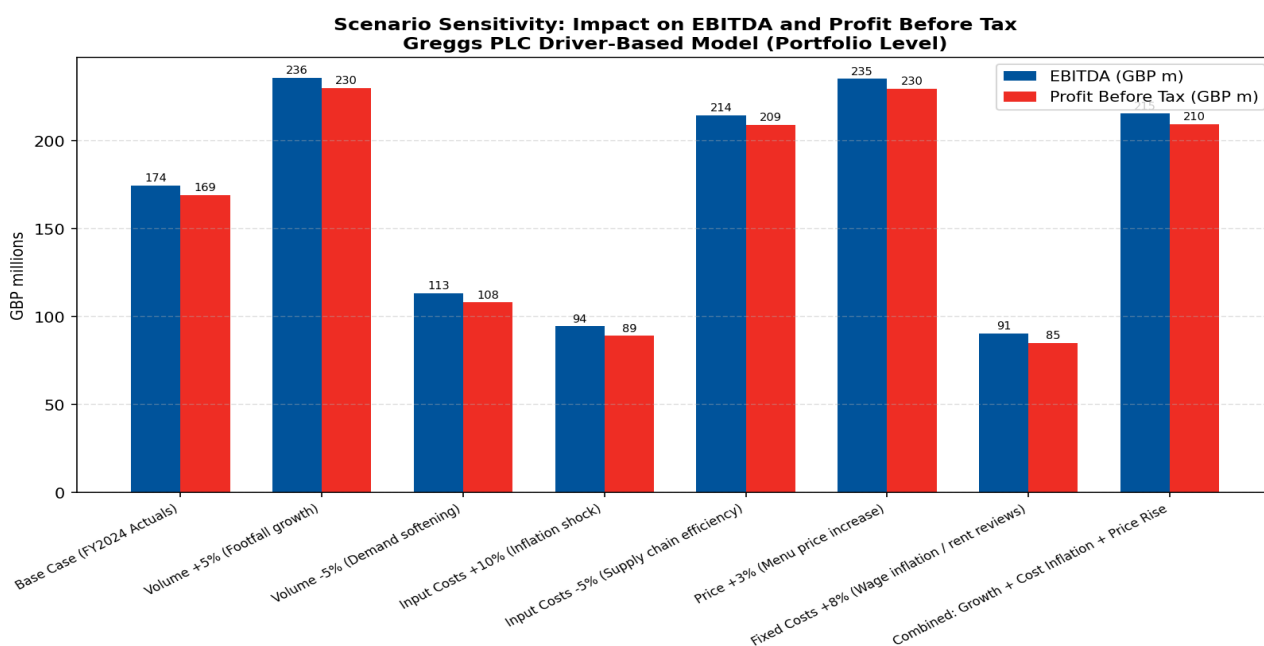


Figure 2: Portfolio-level EBITDA and Profit Before Tax under eight scenarios. Cost-side shocks (input cost inflation, fixed cost growth) have a materially larger downside impact than the equivalent upside from volume or price scenarios – consistent with Greggs' relatively thin EBITDA margin (~8.6% in the base case).

Summary Observations

- **Cost sensitivity dominates:** A 10% rise in input costs has roughly the same absolute EBITDA impact (-£80m) as an 8% rise in fixed costs (-£84m), despite affecting different parts of the cost base – both are first-order risks for the business.
- **Volume and price scenarios are broadly symmetric upside levers:** A 5% volume increase and a 3% price increase each generate a similar EBITDA uplift of approximately £60m, suggesting either lever could be prioritised depending on competitive and demand conditions.
- **Combined scenario remains resilient:** Even when cost inflation (+10%) is modeled alongside growth (+5% volume) and a pricing response (+3%), the combined EBITDA (£215m) and PBT (£210m) both exceed the base case – indicating that a coordinated pricing and growth response can more than offset a significant cost shock.
- **Thin margins amplify volatility:** With a base-case EBITDA margin of ~8.6%, relatively small percentage changes in revenue or cost drivers translate into large percentage swings in EBITDA and PBT – reinforcing the value of continuous scenario monitoring rather than annual-only budgeting.

Prepared by Ayoub Feddal, Financial Analyst – Driver-Based Operational & Financial Simulator project. All figures derived from Greggs plc Annual Report and Accounts 2024 and illustrative driver assumptions as documented in the accompanying README and Colab notebook.